SINGAPORE SCHOOL KINDERLAND

Computer Science — Topic Assessment Worksheet

Topic 1.2.3 & 1.2.4: Data Compression (Lossy, Lossless & RLE) • Total Marks: 25

Student Name: _____ **Date:** _____ **Class:** Sec 2 / O Level

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Question 1: Multiple Choice Questions (5 Marks)

Circle the correct option (i, ii, iii or iv):

1. Which of the following is a primary reason for compressing a file before sending it across the internet?

i. To increase the resolution of the file ii. To reduce transmission time and save bandwidth
iii. To make the file read-only iv. To change analogue data into digital data

2. What happens to original data when a file is compressed using a **Lossy** compression algorithm?

i. All data is preserved and can be restored ii. Redundant data is permanently eliminated
iii. The file size always doubles iv. Data is converted into BCD format

3. MP3 audio compression reduces file size by about 90% mainly by:

i. Removing sounds outside the human hearing range and using perceptual music shaping
ii. Converting audio directly into text strings
iii. Doubling the sample rate
iv. Splitting audio into 8 × 8 pixel blocks

4. JPEG compression separates pixel colour from brightness because the human eye is:

i. More sensitive to colour variations than brightness ii. Less sensitive to colour variations than brightness
iii. Unable to see bright light iv. Only able to see black and white images

5. Which of the following files **MUST** use Lossless compression to prevent system failure?

i. A JPEG holiday photograph ii. An MP3 song track
iii. A software program executable (`game.exe`) iv. An MP4 video clip

Question 2: Classification Table (5 Marks)

Tick (✓) the most appropriate compression type for each scenario below:

No.	Scenario / File Type	Lossy (✓)	Lossless (✓)
i.	Streaming an HD movie over Netflix (`.mp4`)		
ii.	A company financial accounting spreadsheet (`.xlsx`)		
iii.	Uploading a high-resolution photo to social media (`.jpg`)		
iv.	Downloading an operating system setup file (`setup.exe`)		
v.	Listening to music tracks on a smartphone (`.mp3`)		

Question 3: Run-Length Encoding (RLE) Text Calculations (6 Marks)

a) A text document contains the following string of repeated characters:

`X X X X X X  Y Y Y Y  Z Z Z Z Z Z Z Z`

i. State the original uncompressed file size in Bytes (assume 1 Byte per character): [1 mark]
Answer: _____

ii. Show how this string is represented using Run-Length Encoding (RLE): [2 marks]
RLE Representation: _____

iii. Calculate the compressed size in Bytes (assume 2 Bytes per run-length pair) and percentage size reduction: [2 marks]
Compressed Size: _____ Bytes Percentage Reduction: _____ %

b) State why RLE would be ineffective on the string `A B C D E F G H`: [1 mark]
Answer: _____

Question 4: Image Compression & RLE (5 Marks)

An icon bitmap image has an 8 × 8 pixel grid (64 pixels). Each pixel is stored as 1 Byte in monochrome (Black or White). Row 1 and Row 2 of the image are shown below:

- **Row 1:** 8 consecutive Black pixels (`B B B B B B B B`)
- **Row 2:** 2 Black pixels followed by 6 White pixels (`B B W W W W W W`)

i. Calculate the uncompressed size of these two rows in Bytes: [1 mark]
Answer: _____

ii. Write the RLE code for Row 1 and Row 2: [2 marks]
Row 1 RLE: _____ Row 2 RLE:
_____

iii. If each RLE pair takes 2 bytes, calculate the compressed size of these two rows and comment on the saving: [2 marks]
Working & Answer: _____

Question 5: Short Answer Questions & BCD (4 Marks)

a) Explain the difference between **auditory masking** (perceptual shaping) and removing inaudible frequencies in MP3 compression: [2 marks]

_____

_____

b) Convert the decimal value **359.2** into **Binary Coded Decimal (BCD)**: [2 marks]
Working / BCD Nibbles: _____
Final BCD String: _____